

# Lecture 8

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## *Theory of Square Linear System I: Number of Solutions and Matrix Inverse*

### Instructors:

L01 Cosme Louart 程高木

L02 Junfeng Wu 吴均峰

L03&L04 Ruoyu Sun 孙若愚



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

数据科学学院

School of Data Science

# Comments on Course Contents Arrangement

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## *Theme of Lec 3-15: Solving Linear Systems*

(along with Theory of Matrix & Linear Space)

### **Course arrangement:**

We spent Lec 3,4,7 on solving linear systems.

In Lec 7, we learned the final form of using GJ-E to solve square linear systems.

In Lec 8-10, we mainly learn two things:

1) Matrix inverse. (Lec 8 & 10)

This can be used for representing solution of good square linear systems.

2) LU decomposition. (Lec 9)

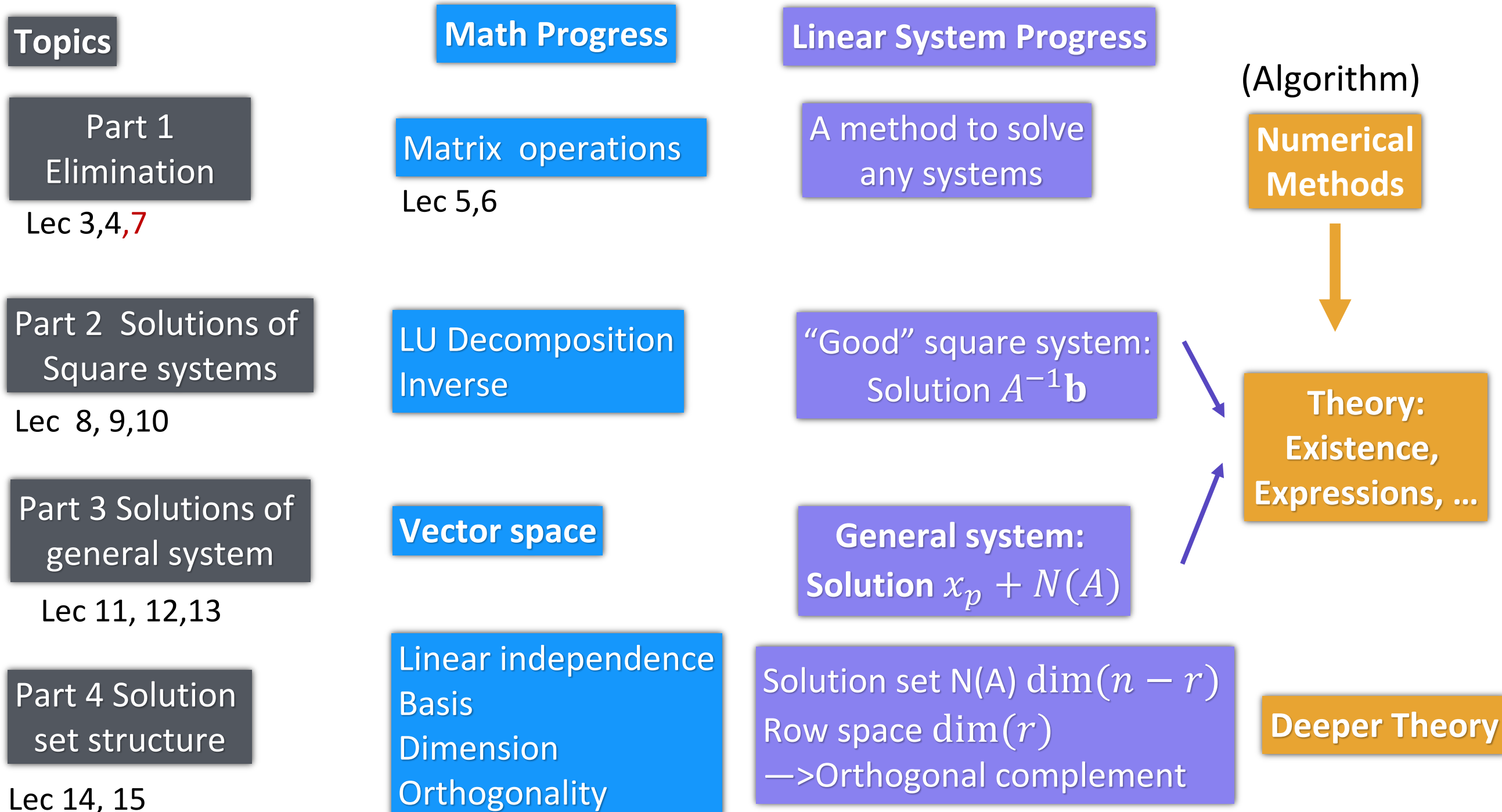
This is a generic result on square matrices.

We will use tools on matrix multiplication from Lec 5-6.

# Roadmap: *Topics Before Midterm*

Lec 1,2: Vector operations.

*Theme of Lec 3-15: Solving Linear Systems* (along with Theory of Matrix & Linear Space)



# Outline and Learning Goals

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Outline:    1) # of Solutions of Linear Systems  
              2) Matrix Inverse

After this lecture, you should be able to

- Tell # of solutions of **square linear systems** by the final RREF
- Apply definition of inverse and properties of inverse
- Prove a few properties of inverse

Strang's book v5: Sec 2.4, 2.5

# Review GJ-E for Square Matrix

[5 mins]

## What if Next Diagonal Entry is 0?

When the next diagonal entry (k,k) is 0, what to do next?

$$\begin{bmatrix} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & & & * \\ 0 & 0 & & & & * \\ 0 & 0 & & & & * \end{bmatrix}$$

# What if Next Diagonal Entry is 0?

When the next diagonal entry (k,k) is 0, what to do next?

$$\left[ \begin{array}{ccccc|c} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & & & * \\ 0 & 0 & & & & * \\ 0 & 0 & & & & * \end{array} \right]$$

**Step 1:** Find a non-zero entry below the next diagonal entry (k,k).

If succeed: Row exchange to make the next diagonal entry nonzero, as pivot.

If no non-zero entry: go to Step 2.

$$\left[ \begin{array}{ccccc|c} 1 & 4 & * & * & * & \\ 0 & 1 & * & * & * & \\ 0 & 0 & 0 & & & \\ 0 & 0 & 3 & & & \\ 0 & 0 & 2 & & & \end{array} \right] \longrightarrow \left[ \begin{array}{ccccc|c} 1 & 4 & * & * & * & \\ 0 & 1 & * & * & * & \\ 0 & 0 & 2 & & & \\ 0 & 0 & 3 & & & \\ 0 & 0 & 0 & & & \end{array} \right]$$

In this example, swap R3 and R5; use (3,3)-th entry 2 as pivot.

Remark: In this example, can swap R3 and R4 as well.

## What if Next Diagonal Entry is 0? 2<sup>nd</sup> Case

**Step 2:** (when all entries below the next diagonal entry is 0)

Check the (k+1)-th column, to find a non-zero entry as pivot

(possibly use row exchange to make sure it is in the k-th row)

In this example: Check the 4<sup>th</sup> column.

Step 2.1: If (4,3)-th entry is nonzero, use it as pivot to eliminate entries below it;  
If (4,3)-th entry is zero, go to Step 2.2.

$$\begin{bmatrix} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * & * \end{bmatrix}$$

# What if Next Diagonal Entry is 0? 3<sup>rd</sup> Case

**Step 2.2:** (when (4,3)-th entry is zero)

Check the 4<sup>th</sup> column; find nonzero entry 2 in 4<sup>th</sup> row.

Swap R3 and R4, to make (4,3)-th entry nonzero, and use it as pivot for elimination.

$$\begin{bmatrix} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 2 & * & * \\ 0 & 0 & 0 & 3 & * & * \end{bmatrix} \xrightarrow{\text{Swap R3 and R4}} \begin{bmatrix} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 2 & * & * \\ 0 & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 3 & * & * \end{bmatrix}$$
$$\xrightarrow{\text{Elimination}} \begin{bmatrix} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 2 & * & * \\ 0 & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * & * \end{bmatrix}$$

## If Still All Zero, Continue...

If in Step 2, no non-zero entries in the  $(k+1)$ -th column below  $(k, k+1)$ ,  
Then check the  $(k+2)$ -th column to find non-zero entries.

In this example: Check the 5<sup>th</sup> column.

$$\begin{bmatrix} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & & * \\ 0 & 0 & 0 & 0 & & * \\ 0 & 0 & 0 & 0 & & * \end{bmatrix}$$

Continue the process.

If cannot find non-zero entry on or below  $k$ -th row in the  
 $k$ -th column,  $(k+1)$ -th column, ...,  $n$ -th column,  
Then everything on or below the  $k$ -th row is 0!!!  
Big zero block at the bottom of the coefficient matrix.

Finish the forward process; begin backward elimination.

## If Still All Zero, Continue...

Continue the process.

If cannot find non-zero entry on or below k-th row in the  
k-th column, (k+1)-th column, ..., n-th column,

$$\left[ \begin{array}{ccccc|c} 1 & 4 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & * \end{array} \right]$$

then everything on or below the k-th row is 0!!!

--Big zero block at the bottom.

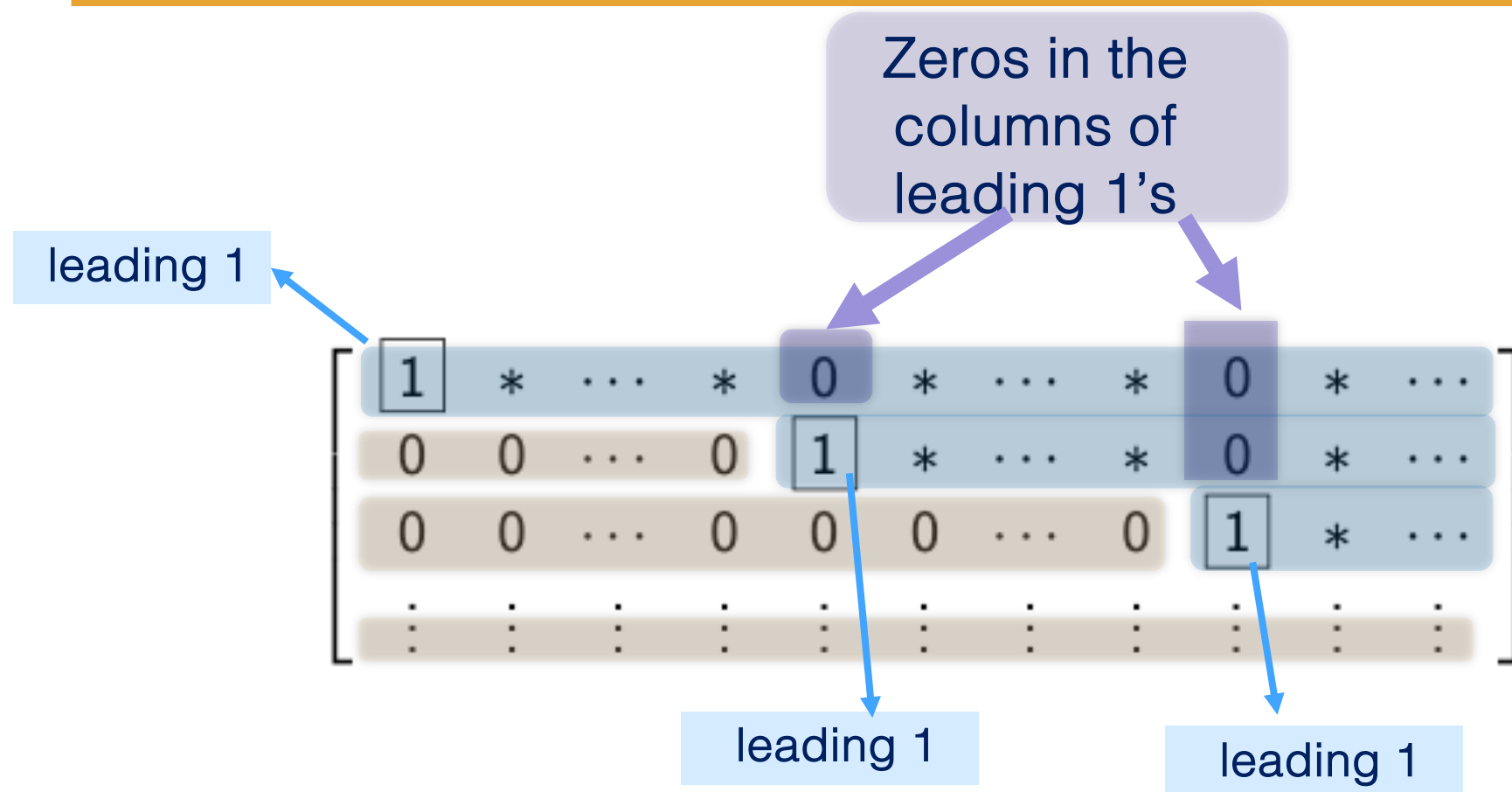
Finish the forward process.

# RREF: One More Condition

## Definition (Reduced Row-Echelon Form)

A matrix is in reduced row echelon form (RREF) if

- i) **Leading 1s:** The first non-zero entry of each non-zero row is 1 (called pivot);
- ii) **Leading 1s Moving Right:** If a row has a leading 1, the leading 1 in the next row appears in a later column;
- iii) **Zero Rows at the Bottom**  
If there is a row whose entries are all zero, then all rows below it have all zero entries.
- iv) **Zeros in the column of the leading 1:** Each leading 1 is the only nonzero entry in its column (the corresponding column is called **pivot column**)



## RREF: Simpler Way to Express

The diagram illustrates a transformation of a matrix from its Reduced Row Echelon Form (RREF) to a form where columns are rearranged. On the left, a 5x5 matrix is shown in RREF. The first three rows have leading ones in columns 1, 2, and 4, respectively. The last two rows are all zeros. The matrix is partitioned into three colored regions: a blue region containing the first three columns, a green region containing the fourth column, and a tan region containing the last two columns. A red arrow points to the right, where the same matrix is shown after column exchange. The columns are now ordered as 1, 2, 3, 4, 5. The first three columns (1, 2, 3) are in the blue region, the fourth column (4) is in the green region, and the last two columns (5, 6) are in the tan region. The leading ones are now in columns 1, 2, and 3.

$$\begin{bmatrix} 1 & 0 & * & 0 & * \\ 0 & 1 & * & 0 & * \\ 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & * & * \\ 0 & 1 & 0 & * & * \\ 0 & 0 & 1 & 0 & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

### Property:

If a matrix  $A$  is in reduced row echelon form (RREF), then exchanging columns of  $A$  can lead to the form

$$\begin{bmatrix} I_k & F \\ 0 & 0 \end{bmatrix}$$

where  $I_k$  is the  $k \times k$  identity matrix.

**Remark:** Standard GJ-E does NOT use column exchange;

Here, we introduce it just for understanding, not for performing GJ-E.

## RREF in the End & Solution Set

**Proposition 7.1** If we solve an  $n \times n$  linear system of equations by Gauss-Jordan Elimination, then we will:  
--obtain REF (which is upper triangular) in the process,  
--obtain RREF in the end.

### Definition (Solution and Solution Set)

A **solution** of an  $m \times n$  linear system with variables  $(x_1, \dots, x_n)$  is a vector  $(s_1, \dots, s_n)$  such that if we let  $x_i = s_i$  for all  $i = 1, \dots, n$

the  $m$  equations hold simultaneously.

The solution set of an  $m \times n$  linear system is a set that contains all solution(s).

# Visualization of Various Cases: 2 by 2 System

- Visualization of a  $2 \times 2$  system of linear equations

$$(i) \quad \begin{aligned} x_1 + x_2 &= 2 \\ x_1 - x_2 &= 2 \end{aligned}$$

Unique solution.

$\sum$  pivots

$$(ii) \quad \begin{aligned} x_1 + x_2 &= 2 \\ x_1 + x_2 &= 1 \end{aligned}$$

No solution.

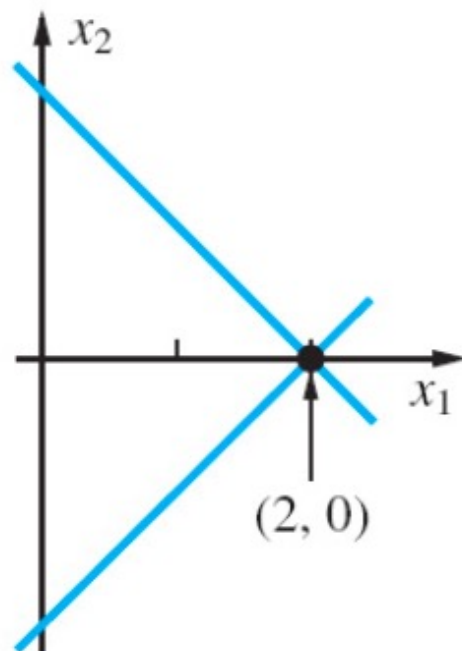
1 pivot

$$(iii) \quad \begin{aligned} x_1 + x_2 &= 2 \\ -x_1 - x_2 &= -2 \end{aligned}$$

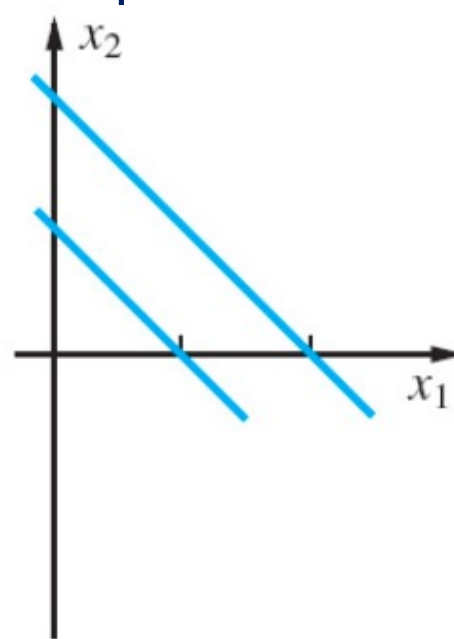
Infinitely many solution.

1 pivot

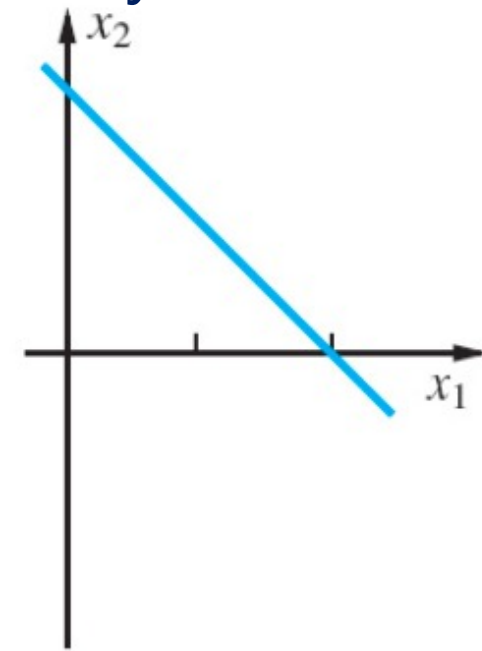
Draw the lines for these equations. What do you observe?



(i)



(ii)



(iii)

Unique intersection.

No intersection.

Two lines overlap:  
Intersects at infinitely  
many points.

# Part 1 Number of Solutions for Square Systems

[10 mins]

## Type 1

primary school.

Augmented matrix  
(RREF & RHS)

$$\left[ \begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array} \right]$$

2 pivots  $\rightarrow$

Equation

$$\begin{cases} x_1 = 1 \\ x_2 = 2 \end{cases}$$

unique solution

$$\left[ \begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{array} \right]$$

$$\begin{cases} x_1 = 1 \\ x_2 = 2 \\ x_3 = 3 \\ x_4 = 5 \end{cases}$$

First type: Unique solution.

--RREF is identity.

--n Pivots

## Type 2

Augmented matrix  
(RREF & RHS)

$$\left[ \begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 0 & 1 \end{array} \right] \quad \text{1 pivot}$$

Equation

$$\begin{cases} x_1 + 2x_2 = 1 \\ 0 = 1 \rightarrow \text{harmful} \end{cases}$$

No solution

$$\left[ \begin{array}{cccc|c} 1 & 0 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0.1 \end{array} \right] \quad \begin{matrix} \text{3 pivots } (< 4) \\ \rightarrow \text{all-zero row} \end{matrix}$$

$$\begin{cases} x_1 + 2x_4 = 3 \\ x_2 = - \\ x_3 = - \\ 0 = 0.1 \end{cases}$$

No solution

Second type: No solution.

--less than n Pivots

--Equation of  $0 = \text{nonzero}$  (contradicting equation)

$$0 = \square \rightarrow \text{nonzero}$$

## Type 3

Augmented matrix  
(RREF & RHS)

$$\left[ \begin{array}{cccc|c} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$< n$  pivots

$\Rightarrow$  all-zero row(s)

$\Rightarrow 0 = \dots$

Equation

$$\begin{cases} x_1 + x_3 + 2x_4 = 1 \\ x_2 + 2x_3 + x_4 = 2 \\ 0 = 0 \\ 0 = 0 \end{cases}$$

$0 = 0 \rightarrow$  useless,  
 $0 = 0 \rightarrow$  harmless

$$\begin{cases} x_1 = 1 - x_3 - 2x_4 \\ x_2 = 2 - x_3 - x_4 \end{cases}$$

$(x_3, x_4) \rightarrow$  free  
determine  $x_1, x_2$

so  $(1 - s - 2t, 2 - s - t, s, t), \forall s, t \in \mathbb{R}$   
is a solution.

Third type: Infinitely many solution.

--less than  $n$  Pivots

--all zero rows correspond to " $0=0$ " equations

# Summary of "Good"/General Square Systems

For solving  $n \times n$  linear system of equations:

**Case 1:**  $n$  pivots.

Unique solution.

"good" case

We may have to exchange the equations (rows).

**Case 2:**  $< n$  pivots.

At the end of elimination, besides  $x_i = \text{***}$  type equation:

Case 2.1: exists  $0 = \text{nonzero number}$  [  $\text{no}$  solution(s) ]

Case 2.2 all are  $0 = 0$  [  $\infty$  many solution(s) ]

This slide will be useful later (e.g., in Lec 10).

# Part II.1 Basics of Inverse

—Definition

—Solution to  $Ax = b$

Strang's book v5: Sec 2.5

[30 mins]

# Motivation

What are the **benefits** of using **notation** of matrix/vector ?

- Simpler;
- Can borrow “rules” of simpler objects

Let's come back to linear systems.

$$ax = b \quad \longleftrightarrow$$

$$A\mathbf{x} = \mathbf{b}$$

What is the solution?

~~$x = \frac{b}{a}$  or  $x = b^{-1}a$   
 $x = ab^{-1}$~~

$x = \frac{b}{a}$  if  $a \neq 0$ . (OK)

If Want a perfect answer, see  
next page

Motivating Question: Can we  
write the solution in a similar  
way, by **creating a notation**?

hope  $x = A^{-1}b$  (show it  
today)

or  $x = \frac{b}{A}$ .

日期: /

Solution of  $ax = b$ ,  $a, b \in \mathbb{R}$

If  $a \neq 0$ ,  $x = \frac{b}{a}$ .

~~Wrong answer~~

~~If  $a = 0$~~

~~no solution.~~

{ If  $a = 0$ ,  $b \neq 0$ , no solution

{ If  $a = 0$ ,  $b = 0$ ,  $\infty$ -many solutions  
( $x \in \mathbb{R}$ ).

# Matrix Inverse

If  ~~$b = \frac{1}{a}$~~ , then  $b = a^{-1}$ .  
 $ab = 1$ .

## Definition (Inverse)

Suppose  $A \in \mathbb{R}^{n \times n}$ . If a matrix  $B \in \mathbb{R}^{n \times n}$  satisfies

$$\underline{AB} = \underline{BA} = \underline{I_n},$$

then we say  $B$  is an inverse of  $A$ ; denoted as  $B = \underline{A^{-1}}$ .

## Definition (Invertible)

Suppose  $A \in \mathbb{R}^{n \times n}$ .

If there exists  $B$  such that  $B = \underline{A^{-1}}$ , then we say  $A$  is invertible. 可逆.

**Questions:** Does the inverse always exist?

# Invertible Matrix

**Question:** Does the inverse always exist?

No! Consider  $n=1$ ; you can find a counter-example.  $A=0 \Rightarrow$  not invertible.

For any  $n$ , you can also find a counter-example (later page).

**Definition** (Singular, nonsingular)

Suppose  $A \in \mathbb{R}^{n \times n}$  (square matrix).

If  $A$  is invertible, then we say  $A$  is nonsingular.

If  $A$  is not invertible, then we say  $A$  is singular.

$n=1$   $n>1$   
 $A \neq 0 \Rightarrow$  nonsingular  $A \neq 0$   
 $A=0 \Rightarrow$  singular  $A=0$

**Remark:** Later, the term “singular” will be extended to non-square matrices.  
But “invertible” is only defined for square matrices.

# Example of Matrix Inverse

**Example** Is the following matrix invertible? If so, find its inverse.

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$$

Solution: Consider a candidate matrix

$$B = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{4} \end{bmatrix}$$

Compute the products:

$$A \cdot B = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 3 \cdot \frac{1}{3} + 0 \cdot 0 & 3 \cdot 0 + 0 \cdot \frac{1}{4} \\ 0 \cdot \frac{1}{3} + 4 \cdot 0 & 0 \cdot 0 + 4 \cdot \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$B \cdot A = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{4} \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \cdot 3 + 0 \cdot 0 & \frac{1}{3} \cdot 0 + 0 \cdot 4 \\ 0 \cdot 3 + \frac{1}{4} \cdot 0 & 0 \cdot 0 + \frac{1}{4} \cdot 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Thus  $A$  is invertible, and  $A^{-1} = B$ .

# Example of Matrix Inverse

## Non-Example

(2) Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$$

Suppose

$$B = \begin{bmatrix} w & x \\ y & z \end{bmatrix} = A^{-1},$$

then

$$\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix} \stackrel{\text{want}}{=} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

But

$$\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix} = \begin{bmatrix} 2w+y & 2x+z \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

In this case,

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \text{ is not invertible.}$$

# Simpler Solution

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$$

Denote

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \vec{v}^T \\ \vec{0}^T \end{bmatrix} \quad \text{where } \vec{v}^T = [2 \quad 1]$$

For any 2 by 2 matrix B, we have

$$AB = \begin{bmatrix} \vec{v}^T \\ \vec{0}^T \end{bmatrix} B = \begin{bmatrix} \vec{v}^T B \\ \vec{0}^T B \end{bmatrix} = \begin{bmatrix} \vec{v}^T B \\ \vec{0}^T \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

# Uniqueness of Matrix Inverse

**Theorem** (Matrix inverse is unique) Suppose the square matrix  $A$  has an inverse. Then  $A^{-1}$  is unique.

Pattern "xxx is unique"

means "If  $X$  satisfies  $P$ ,

$Y$  satisfies  $P$ ,

then  $X = Y$ "

## Example 1

**Statement:** Your personal ID number is unique.

**Equivalent statement:** If  $X$  and  $Y$  are both valid personal ID numbers for you, then  $X=Y$ .

## Example 2

**Statement:** The president of CUHK-SZ is unique.

**Equivalent statement:** If  $A$  is the president of CUHK-SZ, and  $B$  is the president of CUHK-SZ, then  $A=B$ .

# Uniqueness of Matrix Inverse

**Theorem** (Matrix inverse is unique) Suppose the square matrix  $A$  has an inverse. Then  $A^{-1}$  is unique.

**Proof: Need to prove**

If  $AX = XA = I$ ,  $AY = YA = I$ , then  $X = Y$ .

Conditions:

①  $AX = XA = I$

②  $AY = YA = I$

multiply by  $Y$ :  $Y(AX) = Y$ .

$$\begin{aligned} \Rightarrow Y &= Y(AX) = (YA)X \\ &\stackrel{\textcircled{2}}{=} I \cdot X \\ &= X \end{aligned}$$

Remark: Multiplying  $X$  or  $Y$  to four equations lead to 8 relations.

One way that fails: from  $XA = I$ , get  $YXA = \cancel{Y}$

# Uniqueness of Matrix Inverse

**Theorem (Matrix inverse is unique)** Suppose the square matrix  $A$  has an inverse. Then  $A^{-1}$  is unique.

[more compact version of proof]

**Proof.** Let  $B, C$  be the inverse of  $A$ , thus,  $AB = BA = I, AC = CA = I$ .  
Then  $B = BI = BAC = IC = C$

**Wording:**

**Previously:** If  $AB = BA = I_n$ , then we say  $B$  is an inverse of  $A$ .

**Now:** If  $AB = BA = I_n$ , then we say  $B$  is the inverse of  $A$ .

Because of the uniqueness of inverse!

# Invertible Matrix for Solutions of Linear Systems

Recall:  $ax = b$  has a unique solution  $x = \frac{b}{a} = a^{-1}b$  if  $a \neq 0$

Generalization of this result (推广此结果) to linear system:

## Theorem 8.1

If  $A$  is invertible, then  $x = A^{-1}b$  is the unique solution of the linear system  $Ax = b$ .

To understand pattern “xxx is the unique xx of xxx”.

Compare two statements:

**S1: The president of CUHKSZ is unique.**

[true statement]

i.e. If  $A$  is a president of CUHKSZ, and  $B$  is a president of CUHKSZ,  
then  $A=B$ .

**S2: Jim Dai is the unique president of CUHKSZ.** [false statement]

**Meaning:** 1) Jim Dai is “president of CUHKSZ”; [false]

2) If  $A$  and  $B$  are “presidents of CUHKSZ”,  $A$  and  $B$  must be the same.

[true]

Remark: Here, we explain what the statement means, not whether it is true or not.

# Proof: Solutions of Linear Systems with Inverse

## Theorem 8.1

If  $A$  is invertible, then the linear system  $Ax = b$  has a unique solution  $x = A^{-1}b$ .

Proof:

Specify requirements

**Proof requires two things:**

First, prove  $A^{-1}b$  is a solution.

$$A(A^{-1}b) \stackrel{\text{associative}}{=} (AA^{-1})b \stackrel{\text{def of inv}}{=} I \cdot b = b$$

$\Rightarrow A^{-1}b$  is a solution of  $Ax = b$

Second, prove  $A^{-1}b$  is the unique solution; i.e.,

if  $y$  satisfies  $Ay = b$ , then  $y$  must be  $A^{-1}b$

$A^{-1}$   
If  $(Ay) = b$ ,  $\xrightarrow{\text{mult } A^{-1}}$   
then  $(A^{-1}A)y = A^{-1}b$ , then  $y = A^{-1}b$   
 $\stackrel{I \cdot y = y}{\Rightarrow}$   
so  $A^{-1}b$  is the unique solution.

distinguish variable  
v.s. variable with value.

If  $Ay = b$   
then  $y = A^{-1}b$ ,  
then  $x = y$ .

Here  $x$  is symbol/variable.  
 $Ax = b$ ,  $Az = b$ ,  $Au = b$   
 $x$  can be  $z, u, \dots$

# Part III Properties of Inverse

--Inverse of Triangular matrices, products

Strang's book v5 Sec 2.5

[25 mins]

# Matrix Inverse of Special Matrices

## Property 8.1 (Inverse of Diagonal Matrix)

1) A diagonal matrix  $\begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$  is invertible iff

$$a_{ii} \neq 0, \forall i.$$

2) The inverse of the diagonal matrix (if exists) is

$$\begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{a_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{a_{nn}} \end{pmatrix}.$$

# Matrix Inverse of Special Matrices

## Property 8.1 (Inverse of Triangular Matrix)

1) A triangular matrix with diagonal entries  $u_{11}, \dots, u_{nn}$  is invertible iff

$$u_{11} \neq 0, \dots, u_{nn} \neq 0.$$

2) The inverse of an upper triangular matrix  
(if exists) is upper triangular matrix.

A. upper  
B. lower  
C. none.

3) The inverse of a lower triangular matrix  
(if exists) is lower triangular matrix.

Will be useful later!

# Matrix Inverse of Lower Triangular Matrices (1)

Use  $n=2$  as illustration.

Suppose  $A = \begin{bmatrix} a & 0 \\ c & b \end{bmatrix}$ . Under what conditions,  $A^{-1}$  exists?

One direction: Find out the condition, assuming its inverse exists. (identify necessary condition)

We assume  $A^{-1}$  has the form:  $A^{-1} = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$

$$AA^{-1} = I \Rightarrow \begin{bmatrix} a & 0 \\ c & b \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} ax & ay \\ cx + bz & cy + bw \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Assume  $ax=1$ , then?

$\exists x, y, z, w$  s.t.

- ①  $ax = 1 \Rightarrow x = \frac{1}{a}$  ⑤ &  $a \neq 0$  ⑥
- ②  $ay = 0 \Rightarrow y = 0$  ⑦
- ③  $cx + bz = 0 \xrightarrow{⑤} \frac{1}{a} \cdot c + bz = 0 \Rightarrow z = -\frac{c}{ab}$  ⑧
- ④  $cy + bw = 1 \xrightarrow{⑦} bw = 1 \Rightarrow w = \frac{1}{b}$  ⑨ &  $b \neq 0$  ⑩

Conclusion: (1) if  $A^{-1}$  exists, we must have  $a \neq 0, b \neq 0$ .

(2) Expression of  $A^{-1}$  can only be  $\begin{bmatrix} 1/a & 0 \\ -c/ab & 1/b \end{bmatrix}$  upper triangular (lower)

## Matrix Inverse of Lower Triangular Matrices (2)

Suppose  $A = \begin{bmatrix} a & 0 \\ c & b \end{bmatrix}$ . Under what conditions,  $A^{-1}$  exists?

Sufficient

The other direction: Prove "If  $a \neq 0, b \neq 0$ , then  $A^{-1}$  exists".

Conclude  
Define  $B = \begin{bmatrix} \frac{1}{a} & 0 \\ -\frac{c}{ab} & \frac{1}{b} \end{bmatrix}$

(last page)

Verify:  $AB = \begin{bmatrix} a & 0 \\ c & b \end{bmatrix} \begin{bmatrix} \frac{1}{a} & 0 \\ -\frac{c}{ab} & \frac{1}{b} \end{bmatrix} = \dots = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$BA = \begin{bmatrix} \frac{1}{a} & 0 \\ -\frac{c}{ab} & \frac{1}{b} \end{bmatrix} \begin{bmatrix} a & 0 \\ c & b \end{bmatrix} = \dots = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Conclusion: B is the inverse of A.

Final conclusion:  $\begin{bmatrix} a & 0 \\ c & b \end{bmatrix}$  is invertible iff  $a \neq 0, b \neq 0$

# Properties of Inverse: Inverse of Products

**Theorem (Matrix Inverse of Matrices Product)** Suppose  $A$  and  $B$  are invertible matrices of size  $n$ , then  $AB$  is invertible and

Candidate  $(AB)^{-1} = B^{-1}A^{-1}$

Proof: Let  $C = B^{-1}A^{-1}$ . Need to show  $(AB)^{-1} = C$ .  
By definition, need to show  $AA^{-1} = I$  and  $BB^{-1} = I$ . } goal

The LHS of these equations are:

$$C(AB) = (B^{-1}A^{-1}) \cdot AB = B^{-1}(\underbrace{A^{-1}A}_I)B = B^{-1}B = I$$

$$(AB)C = (AB) \cdot (B^{-1}A^{-1}) = A(\underbrace{B \cdot B^{-1}}_I)A^{-1} = AA^{-1} = I$$

This means  $C = (AB)^{-1}$  by def.

# Inverse Exercise

## Exercise

$A_1, A_2, \dots, A_k$  are real invertible matrices.

Question: What is  $(A_1 A_2 \dots A_k)^{-1}$ ?

$$A_k^{-1} A_{k-1}^{-1} \dots A_1^{-1}$$

Product of invertible matrices is invertible.

Will be useful later this lecture!

# Inverse Exercise

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Judgement: The product of any  $n \times n$  invertible matrices is invertible.



# Concluding Section



# Summary Today

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## One sentence summary:

We learned

## Detailed summary:

# Summary Today Instructor's summary

## One sentence summary:

We learned # of solutions of square linear system based on RREF,  
inverse: def & properties

## Detailed summary:

- 1) # of solutions of square  $n \times n$  system.  
Case 1:  $n$  pivots. unique solution.  
Case 2:  $< n$  pivots.  $\left\{ \begin{array}{l} \text{Case 2.1. } 0 = \text{nonzero: no solution} \\ \text{Case 2.2. } 0 = 0: \infty\text{-many solutions.} \end{array} \right.$
- 2) Definition:  $B = A^{-1}$  if  $AB = I$  &  $BA = I$ .
- 3) Properties:
  - i) inverse is unique;
  - ii) When  $A^{-1} \exists$ ,  $Ax = b$  has unique solution  $x = A^{-1}b$ ;
  - iii) When  $A^{-1}, B^{-1} \exists$ ,  $(AB)^{-1} = B^{-1}A^{-1}$ ;
  - iv) Triangular matrix  $U$  inverse:
    - a)  $U^{-1} \exists$  iff diag entries  $\neq 0$ ;
    - b)  $U^{-1}$  some shape.  $U \rightarrow U$   
 $L \rightarrow L$ .